Revealing the Modular Structure of Collatz Trajectories through Binary Logarithmic Spiral Plots:

A Topological Analysis of 4m+1 Magazines and $a \cdot 2^b$ Launchers

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Abstract

This study proposes a visualization method that reveals structural patterns hidden within the seemingly chaotic odd-number transitions of the Collatz map by embedding its trajectories into a base-2 logarithmic spiral coordinate system. In this framework, sequences generated by the recurrence $4m+1$ are treated as "magazines," while even-number sequences of the form $a \cdot 2^b$, rooted at an odd integer a, are treated as "launchers." These objects appear as geometric lanes with distinct angular directions on the logarithmic spiral plane, allowing the transitions of Collatz trajectories between these lanes to be visually tracked.

Using the trajectory of $n = 7$ as a representative example, we demonstrate how the orbit progresses through multiple magazines and launchers in a stepwise manner toward convergence.

Trajectory for $n = 7$: 7 → 11 → 17 → 13 → 5 → 1

Although this work does not aim to prove the Collatz conjecture, it provides a new visual framework for understanding the structural features of Collatz trajectories.

1. Introduction

The Collatz conjecture (the $3n+1$ problem) remains unsolved despite the simplicity of its defining rules, largely due to the complex behavior exhibited by its trajectories. Previous studies have focused primarily on linear analyses such as step counts or behavior along the number line, but the geometric "shape" of trajectories has not been sufficiently explored.

In this study, any natural number n is mapped into the following polar coordinate system:

$$r = \log_2(n)$$

$$\theta = 2\pi \cdot \log_2(n)$$

This mapping separates two aspects of the number:

r: magnitude (scale)

θ: angular phase relative to powers of 2

This enables Collatz trajectories to be visualized as "jumps" between lanes with specific angular directions.

2. Theoretical Framework

The behavior of odd numbers in the Collatz map is organized into three modules. These names are introduced for explanatory convenience and correspond to mathematical classifications.

2.1. Odd numbers of the form $4k+1$ (Cooling Odd Numbers)

An odd number $n = 4k+1$ transforms as follows:

$$3n + 1 = 12k + 4 = 4(3k + 1)$$

Thus, at least two divisions by 2 are guaranteed. On the logarithmic spiral plane, this appears as a deterministic inward contraction.

The sequence generated by the following recurrence is referred to as a "$4m+1$ magazine":

$$n_{(k+1)} = 4n_{(k)} + 1$$

2.2. Odd numbers of the form *4k+3* (Heating Odd Numbers)

An odd number $n = 4k+3$ transforms as:

$$3n + 1 = 12k + 10 = 2(6k + 5)$$

After a single division, it immediately transitions to a new odd number. On the logarithmic spiral plane, this appears as an outward jump accompanied by a large angular rotation.

2.3. Even sequences $a \cdot 2^b$ (Glide Launchers)

For any odd integer a, the following even sequence forms a straight radial descent on the logarithmic spiral plane, maintaining a constant angular phase:

$$a, 2a, 4a, 8a, \ldots.$$

This structure is referred to as a "launcher."

3. Trajectory Analysis

This section analyzes the trajectory of $n = 7$ by tracing it backward from its endpoint (**Figure 1**).

Trajectory for $n = 7$: $7 \rightarrow 11 \rightarrow 17 \rightarrow 13 \rightarrow 5 \rightarrow 1$

At each step, we identify the magazine to which the odd number belongs and the launcher into which it transitions.

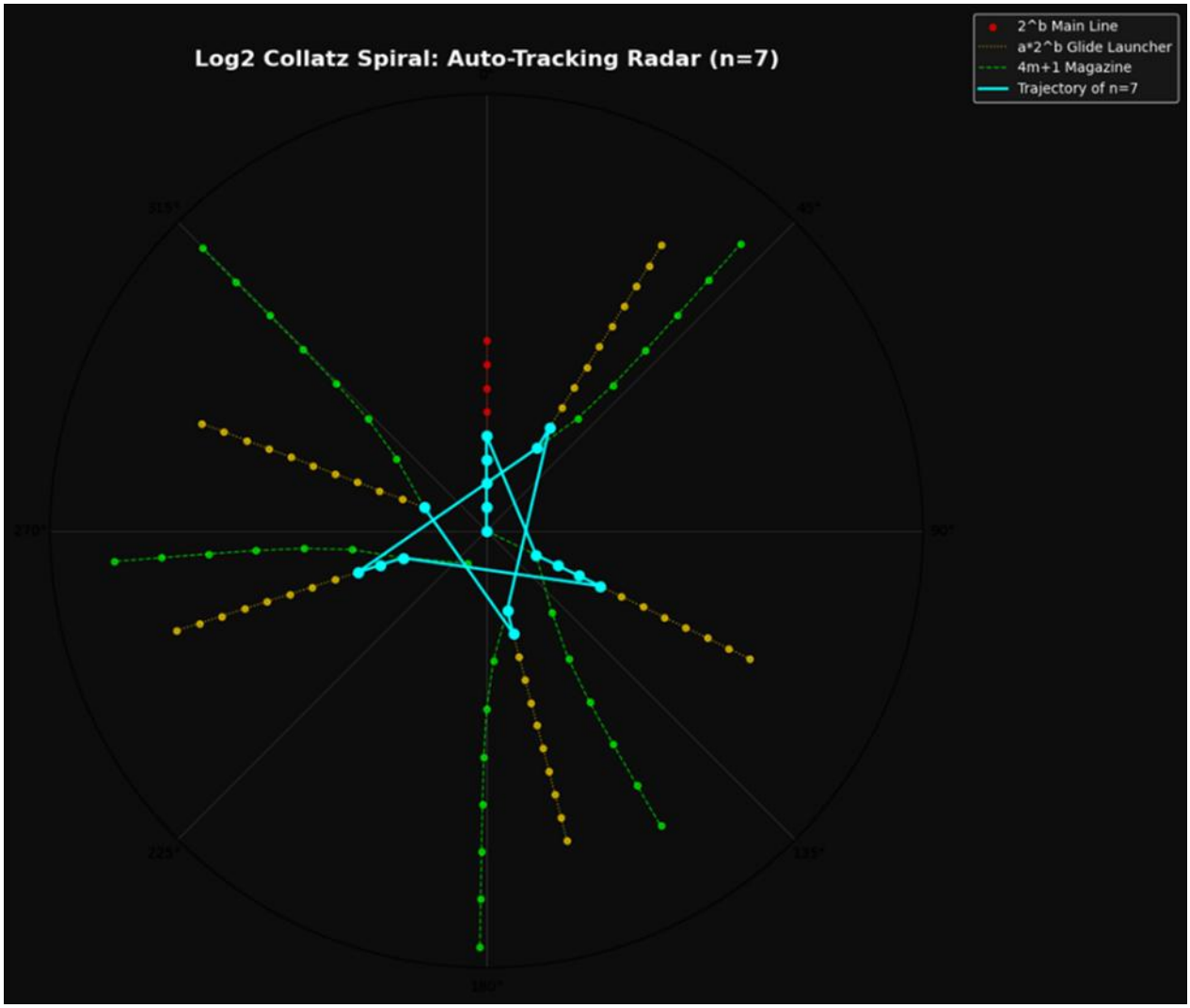

Figure 1. Log_2 Collatz spiral auto-tracking radar (initial value $n = 7$).

3.1. Phase 1: Jump to the final 2^b lane

The 1-series magazine (1, 5, 21, …) contains the Cooling Odd Number 5. Applying $3n+1$ yields 16 (= 2^4), placing the trajectory onto the final 2^b lane. From here, four radial steps lead directly to 1. Other members of the 1-series magazine behave similarly (**Figure 2**).

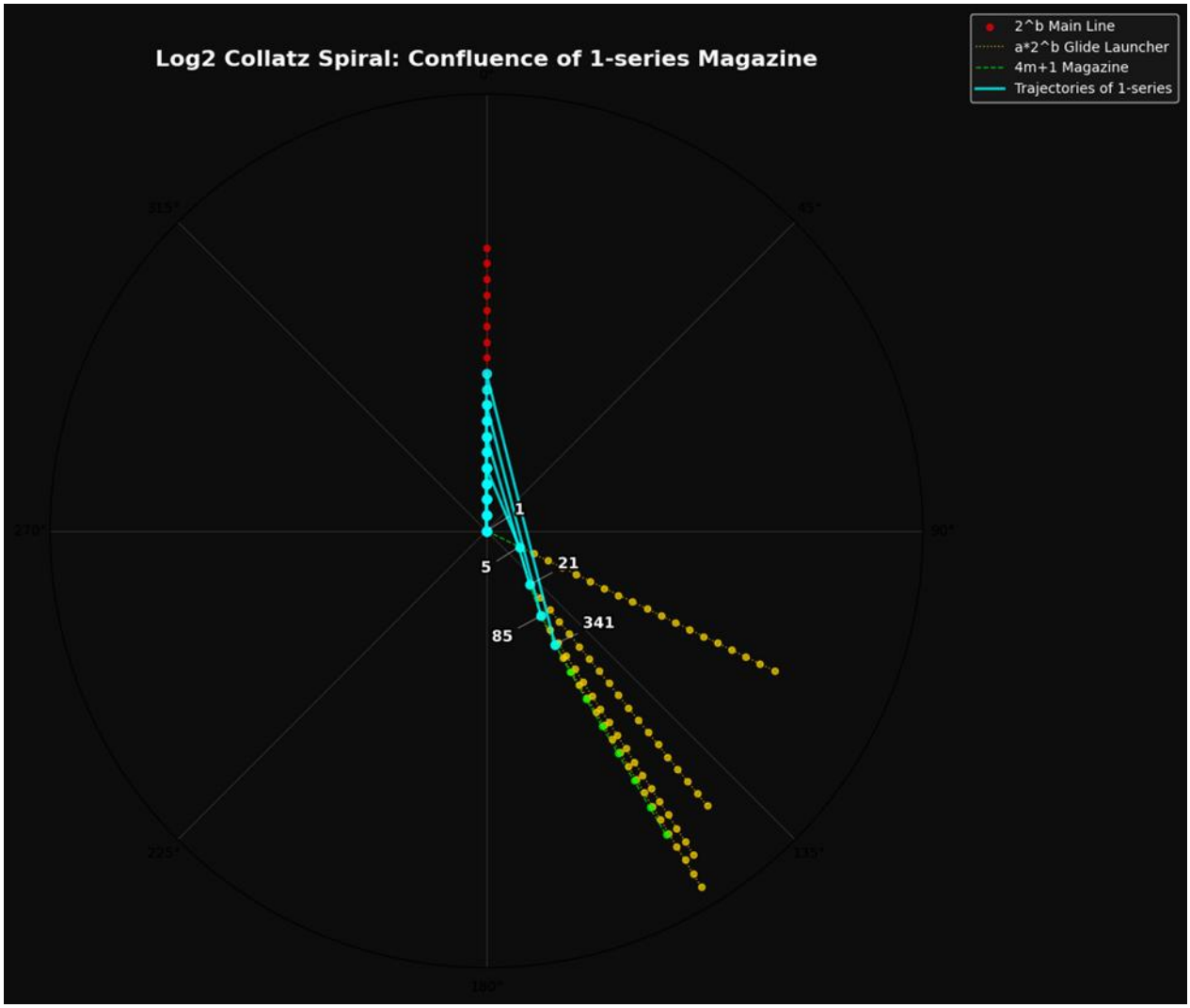

Figure 2. Confluence of the 1-series magazine into the 2^b axis of the final lane.

3.2. Phase 2: Convergence into the $5{\cdot}2^b$ launcher

The 3-series magazine (3, 13, 53, …) contains the Cooling Odd Number 13, which transforms as:

$$3{\cdot}13 + 1 = 40 \ (= 5{\cdot}2^3)$$

Thus, the trajectory enters the $5{\cdot}2^b$ launcher and descends three steps to reach 5. Other members of the 3-series magazine behave similarly (**Figure 3**).

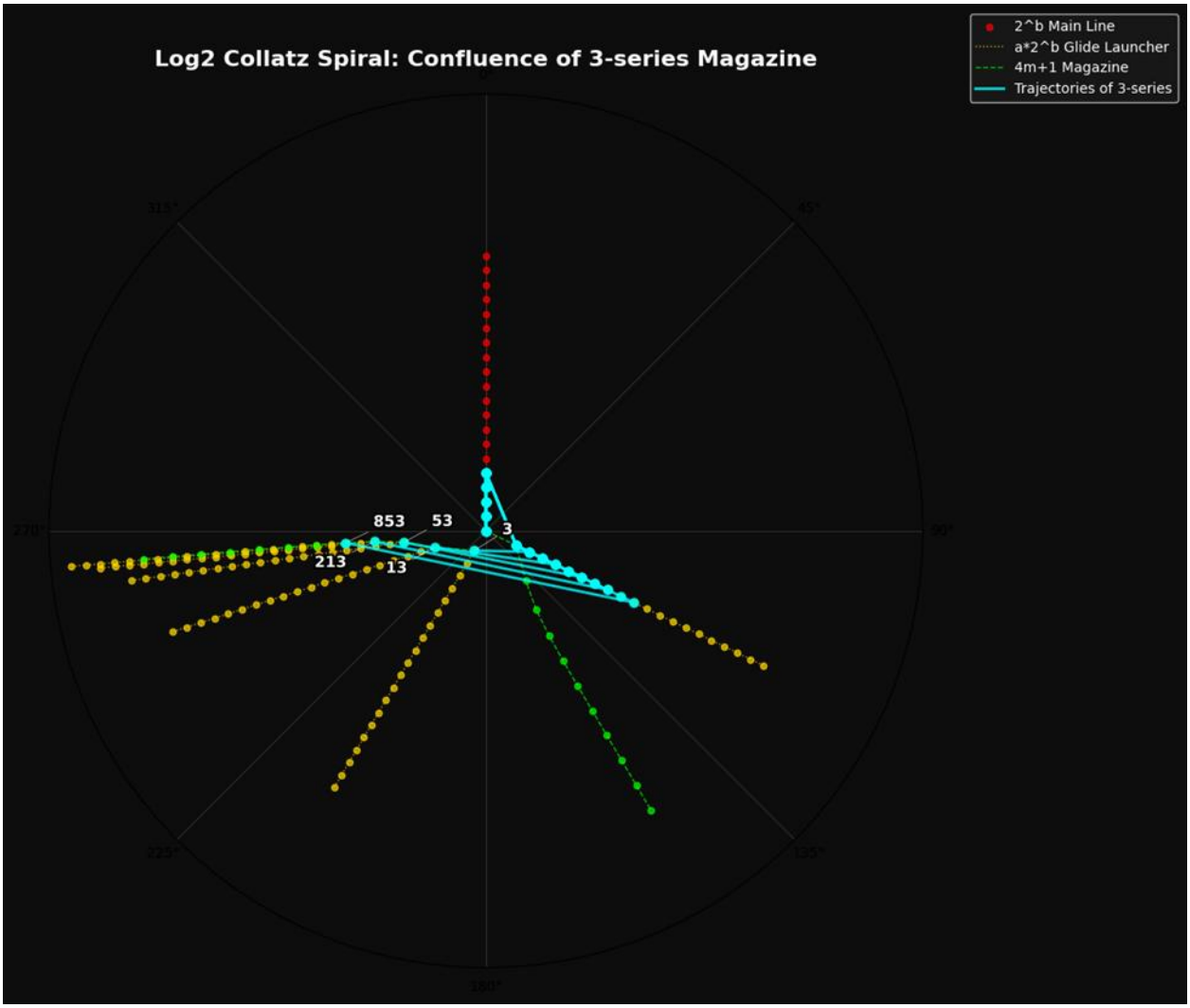

Figure 3. Confluence of the 3-series magazine into the $5{\cdot}2^b$ launcher.

3.3. Phase 3: Convergence into the $13 \cdot 2^b$ launcher

The 17-series magazine (17, 69, 277, …) contains the Cooling Odd Number 17, which transforms as:

$$3 \cdot 17 + 1 = 52 \; (= 13 \cdot 2^2)$$

The trajectory enters the $13 \cdot 2^b$ launcher and descends two steps to reach 13. Other members of the 17-series magazine behave similarly (**Figure 4**).

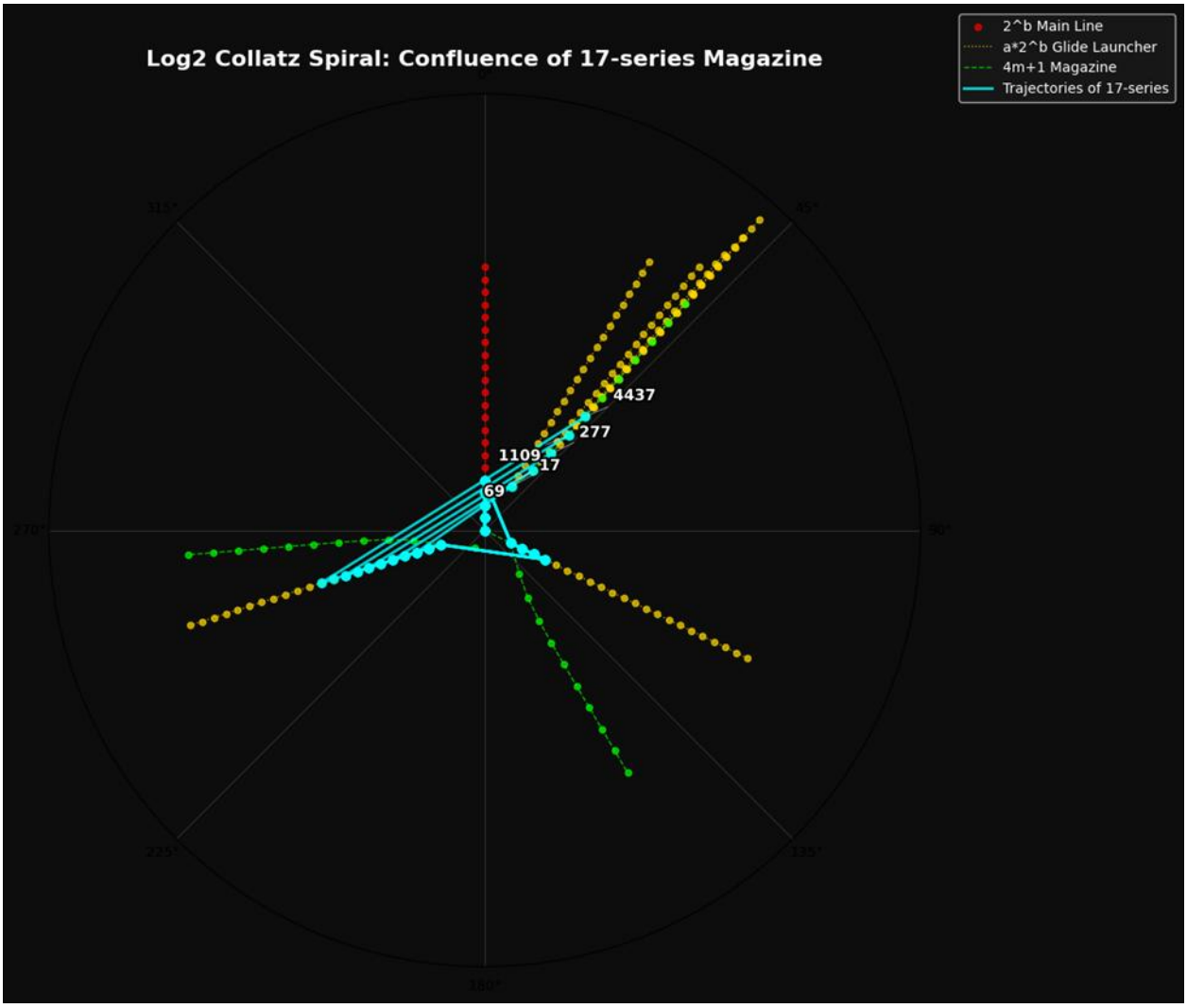

Figure 4. Confluence of the 17-series magazine into the $13 \cdot 2^b$ launcher.

3.4. Phase 4: Convergence into the $17 \cdot 2^b$ launcher

The 11-series magazine (11, 45, 181, …) contains the Heating Odd Number 11, which transforms as:

$$3 \cdot 11 + 1 = 34 \ (= 17 \cdot 2^1)$$

The trajectory enters the $17 \cdot 2^b$ launcher and descends one step to reach 17. Other members of the 11-series magazine behave similarly (**Figure 5**).

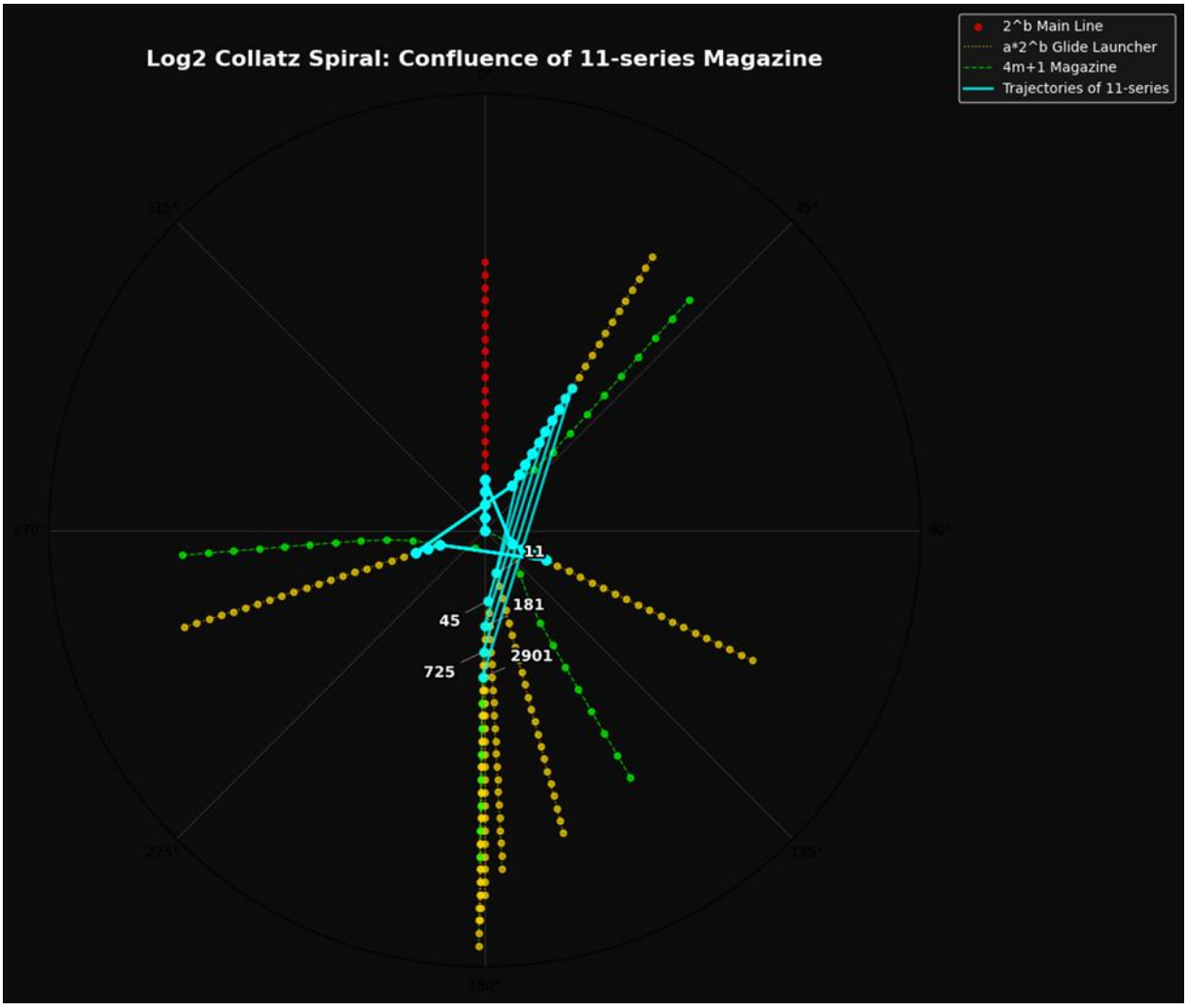

Figure 5. Confluence of the 11-series magazine into the $17 \cdot 2^b$ launcher.

3.5. Phase 5: Convergence into the $11 \cdot 2^b$ launcher

The 7-series magazine (7, 29, 117, …) contains the Heating Odd Number 7, which transforms as:

$$3 \cdot 7 + 1 = 22 \ (= 11 \cdot 2^1)$$

The trajectory enters the $11 \cdot 2^b$ launcher and descends one step to reach 11. Other members of the 7-series magazine behave similarly (**Figure 6**).

From these observations, the trajectory of $n = 7$ can be described as the deterministic sequence:

Heating (7) → Heating (11) → Cooling (17) → Cooling (13) → Cooling (5) → Convergence (1)

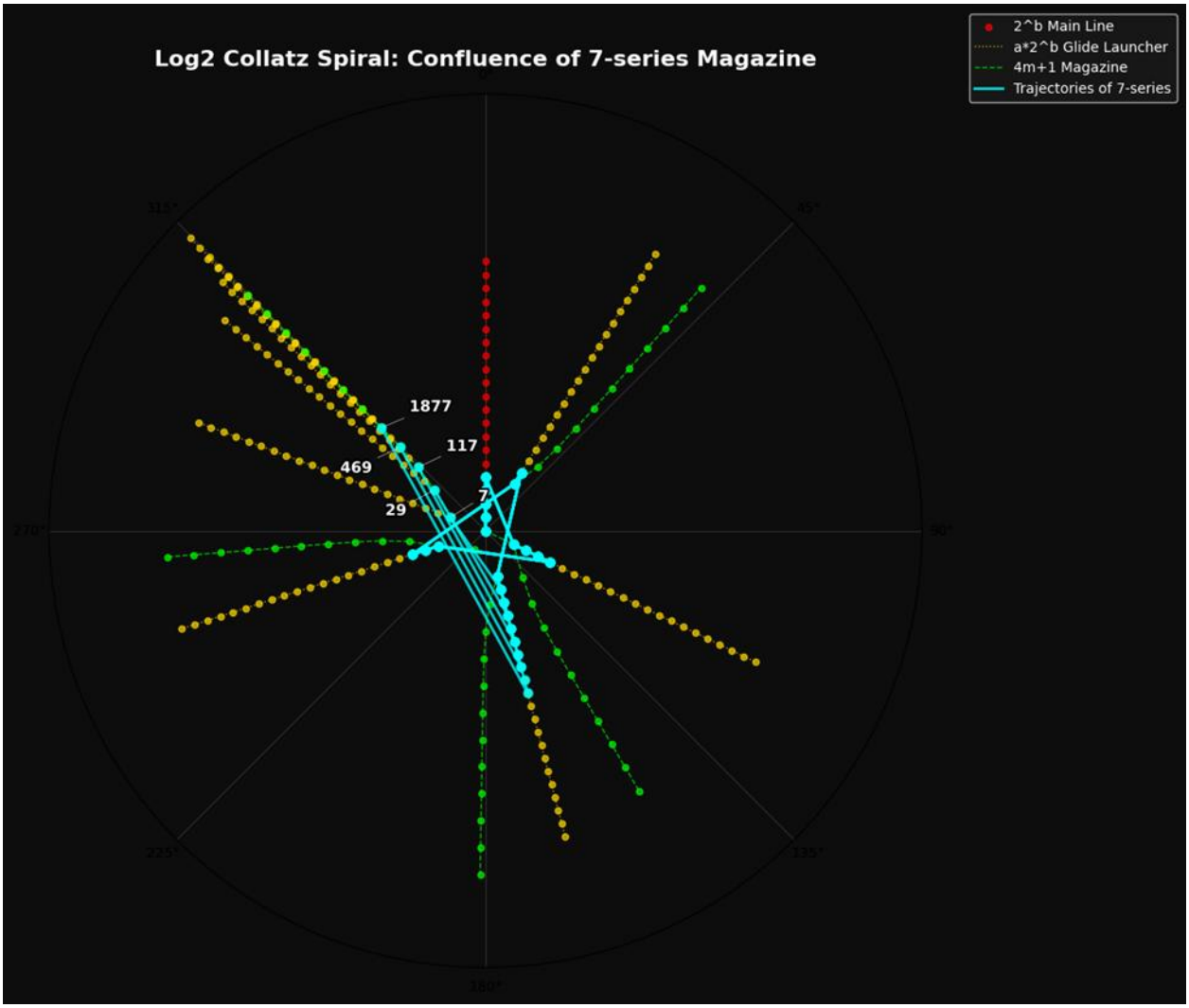

Figure 6. Confluence of the 7-series magazine into the $11 \cdot 2^b$ launcher.

4. Discussion

Visualization on the binary logarithmic spiral plane reveals that Collatz trajectories follow a recurring asymmetric cycle:

- outward jumps generated by odd numbers of the form $4k+3$ (Heating Odd Numbers)
- phase-fixed radial descent along $a \cdot 2^b$ launchers
- deterministic contraction generated by odd numbers of the form $4k+1$ (Cooling Odd Numbers)

This cycle provides a useful structural perspective on Collatz dynamics.

5. Conclusion

By combining the binary logarithmic spiral plot with an auto-tracking radar algorithm, this study presents a method for extracting and visualizing structural features of Collatz trajectories. While not intended as a proof of the Collatz conjecture, the method offers a new geometric viewpoint for understanding the behavior of Collatz orbits.

6. Appendix

Visualization code (MIT License)

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The accompanying source code is released under the MIT License.

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